

Software architecture

A request from the browser to evidence and back

1. Browser

Public pages request /api/data. A user signs in through the platform-owned ChatGPT flow, then explicitly registers for this application.

2. Sites backend

Every write checks the request origin. AI requires identity plus a users record. Refresh needs editor/admin. Saving assumptions and role changes need admin. Identity comes from the trusted Sites dispatcher, never from a user-entered ID.

3. Cloudflare D1

Sources, metrics, countries, designs, claims and registrations persist in eight tables including audit events and rate limits. Prepared statements bind values. Schema migrations run before deployment. An idempotent seed batch inserts the initial evidence separately.

4. Deterministic model

The same calculation functions power the browser and adviser tools. They calculate site power, productive GPU-hours, ten-year costs and three stress cases. The language model explains these results instead of inventing arithmetic.

5. Approved external API

An editor can refresh a fixed World Bank endpoint. The backend validates country, indicator, year, range and numeric values before committing a batch. Any failure preserves the last valid observations. Each record retains its retrieval date.

6. OpenAI Responses API

After registration and hourly rate checks, the server retrieves relevant D1 evidence and calls OpenAI with controlled read/calculation tools. At most three model turns per question. Server credentials never reach the browser. Sources are data, not instructions.

7. Cited response

The server validates JSON structure and source IDs against retrieved records, then returns the answer, assumptions, uncertainty, links and token count. Invalid citations fail closed. Source-ID validity does not by itself prove that a source entails the claim.

Operational boundary

The native platform provides hosted identity. Local preview uses a loopback-only mock user. Live AI answers and injection resistance require an approved hosted key and real testing. Initial publication is private, so anonymous hosted access remains unverified until the owner authorizes a public audience.